

Infection Control Study Guide (Chapter 10)

Focused Review for NBDHE Preparation (Questions 1 to 51 & Cases A & B)

How to use this guide: Study the yellow "High-Yield" callout boxes first. They contain the core infection control facts tested directly in Questions 1 to 51 and Cases A & B. Then use the question-to-concept map, clinical vocabulary table, and rapid recall summary at the end to master your terminology.

1. Regulatory Agencies, Guidelines & Professional Bodies

- **OSHA (Occupational Safety and Health Administration):**
 - Division of the US Department of Labor. **Federal and state regulatory enforcement body.**
 - Authority to formulate and **enforce workplace safety regulations** (Bloodborne Pathogens Standard, hazard communication, PPE, sharps safety).
 - Requires employers to provide post-exposure evaluation and follow-up at **no cost to employees**. Does NOT mandate paid time off or pay for medical treatment (handled via Worker's Compensation).
- **CDC (Centers for Disease Control and Prevention):**
 - Advisory public health agency under DHHS. Formulates **evidence-based recommendations** and guidelines for infection prevention in healthcare settings (not an enforcement agency).
- **FDA (Food and Drug Administration):**
 - Regulates the safety and efficacy of **medical devices**, sterilizers, ultrasonic cleaners, gloves, surgical masks, and chemical sterilants/disinfectants.
- **EPA (Environmental Protection Agency):**
 - Regulates hospital disinfectants and environmental waste management.
- **ADS (Association for Dental Safety, formerly OSAP):**
 - Promotes science-based dental infection control policies and safety education.

[!NOTE] **HIGH-YIELD FOR QUESTIONS 1, 2, 33, 39, 41, 42:** * **OSHA** is a **regulatory enforcement body** that enforces workplace safety laws. * The **CDC** formulates **evidence-based recommendations** for disease prevention. * The **FDA** regulates the safety and effectiveness of **sterilizers and ultrasonic units**. * Employers must provide post-exposure medical follow-up at **no cost to the employee**.

2. Chain of Infection & Transmission Pathways

- **Elements of Chain of Infection:** Pathogen (source) → Reservoir → Portal of exit → Mode of transmission → Portal of entry → Susceptible host.
- **Pathways of Disease Transmission:**
 - **Direct Contact:** Person-to-person physical exposure of nonintact skin or mucous membranes to infected blood or bodily fluids (e.g., bare hands touching blood).
 - **Indirect Contact:** Contact with contaminated inanimate objects or instruments (e.g., **sharps injury with contaminated explorer/scaler**, touching contaminated surface).

- **Droplet Transmission:** Exposure of mucous membranes (eyes, nose, mouth) to larger spatter or droplets generated during coughing, sneezing, or ultrasonic scaling at close range (within 14–18 inches).
- **Airborne Infection:** Inhalation of microscopic aerosolized droplets or water vapor suspended in air (e.g., aspirating *Legionella* water vapor from DUWL or *Mycobacterium tuberculosis*).

[!NOTE] **HIGH-YIELD FOR QUESTIONS 7, 9, 10, 11, 12, 15, 39, 40 & CASE STUDY A:** * A sneeze or cough represents a **portal of exit** (and portal of entry into a new host). * **Indirect contact** involves exposure via contaminated instruments or sharps injuries. * **Airborne infection** occurs via inhalation of aerosolized particles (e.g., *Legionella* in water vapor). * **Standard Precautions** treat blood, **all body fluids (except sweat)**, nonintact skin, and mucous membranes as potentially infectious. * Patients with **active tuberculosis** must be treated in an **airborne isolation room** (not a standard dental operatory).

3. Spaulding Classification & Instrument Processing Protocols

- **Spaulding Classification System:**
- **Critical Items:** Penetrate soft tissue, bone, or bloodstream (e.g., **dental hygiene cures, scalers, extraction forceps, surgical burs**). Must be **heat sterilized**.
- **Semicritical Items:** Touch mucous membranes or nonintact skin without penetrating soft tissue or bone (e.g., **mouth mirrors, impression trays, radiograph sensors/film holders**). Must be heat sterilized or processed with high-level disinfectant.
- **Noncritical Items:** Touch intact skin only (e.g., **blood pressure cuffs, X-ray tube heads, pulse oximeters**). Processed with low- to intermediate-level disinfectant.
- **Cleaning & Sterilization Protocols:**
- **Pre-cleaning Bioburden:** Automated cleaning (ultrasonic bath, instrument washer) is preferred over hand scrubbing. If hand scrubbing is required, use a long-handled brush (short-handled brush is an EXCEPTION/improper technique).
- **Wicking:** Wet sterile packages can draw environmental microbes into the package via capillary action (**wicking**). Instruments **MUST** be completely dry before packaging/sterilization.

[!NOTE] **HIGH-YIELD FOR QUESTIONS 4, 5, 6, 16, 27, 44:** * **Critical items** (cures, scalers, forceps) penetrate soft tissue and **must be heat sterilized**. * **Semicritical items** (mouth mirrors, X-ray sensors) touch mucous membranes. * **Noncritical items** (blood pressure cuff) touch intact skin only. * Hand scrubbing with a short-handled brush is an **unacceptable cleaning method**. * **Wicking** occurs when wet instrument packages absorb microbial contamination.

4. Environmental Surface Disinfection & Controls

- **Environmental Surface Categories:**
- **Clinical Contact Surfaces:** High-touch surfaces in patient care area (light handles, switches, air/water syringe buttons, handpiece tubing, computer keyboards). Must be protected with surface barriers or cleaned/disinfected between patients with an **intermediate-level disinfectant**.
- **Housekeeping Surfaces:** Floors, walls, sinks. Cleaned regularly with soap/water or low-level disinfectant.
- **Disinfectant Levels & Kill Claims:**

- **High-Level Disinfectant:** Kills all micro-organisms including bacterial endospores (used as chemical sterilant for heat-sensitive semicritical items, e.g., glutaraldehyde).
- **Intermediate-Level Disinfectant:** Hospital disinfectant with a **tuberculocidal claim** (kills *Mycobacterium tuberculosis*, HBV, HIV, *S. aureus*). Does NOT kill *C. difficile* spores.
- **Low-Level Disinfectant:** Hospital disinfectant without tuberculocidal claim.
- **Engineering vs. Work Practice Controls:**
- **Engineering Controls:** Technology or devices that isolate/remove bloodborne hazards (e.g., **sharps containers, self-sheathing needles, needle recapping devices**).
- **Work Practice Controls:** Behaviors/actions that reduce risk (e.g., **one-handed scoop recapping technique**, maintaining working distance, fulcruming safely).

[!NOTE] **HIGH-YIELD FOR QUESTIONS 9, 10, 19, 20, 21, 23:** * **Engineering controls** are devices (e.g., sharps containers, self-sheathing needles) that isolate hazards. * **One-handed scoop technique** is a **work practice control**. * **Intermediate-level disinfectants** must carry a **tuberculocidal kill claim**; they do NOT kill *C. difficile*. * Housekeeping surfaces (floors) are NOT clinical contact surfaces.

5. Dental Unit Waterlines (DUWL) & Legionella (Case Study A)

- **DUWL Biofilm Formation:** Stagnant water in narrow DUWL promotes microbial biofilm accumulation containing opportunistic pathogens like *Legionella pneumophila* and *Pseudomonas aeruginosa*.
- **Legionnaires' Disease (Case Study A):**
- **Case Scenario:** An elderly patient contracts Legionnaires' disease after dental treatment in an operatory without treated waterlines.
- **Pathway of Transmission:** **Community-to-patient** via airborne inhalation of contaminated water aerosols/vapor.
- **Portal of Entry:** Inhalation of water vapor into lungs (**airborne infection**).
- **Prevention:** Dental unit waterlines must be treated and tested to ensure output water contains **<= 500 CFU/mL** of heterotrophic mesophilic water bacteria.

[!NOTE] **HIGH-YIELD FOR CASE A (QUESTIONS 1, 2, 3, 4, 5, 6):** * Legionnaires' disease in Case A resulted from **community-to-patient** transmission via **untreated dental unit waterlines**. * The portal of entry was **inhalation of aerosolized water vapor** into the lungs. * *Legionella* is a **bacterium** naturally present in municipal water. * Waterline treatment ensures water quality stays **<= 500 CFU/mL**.

6. Occupational Exposure Management, Immunizations & Work Restrictions (Case Study B)

- **Post-Exposure Sharps Protocol (Case Study B):**
- **Immediate First Aid:** Immediately **wash the punctured area thoroughly with soap and water**. (Do NOT scrub with a brush, squeeze the wound, or apply bleach).
- **Reporting & Follow-Up:** Report to Infection Control Coordinator. Employer must immediately arrange and pay for confidential medical evaluation, source patient testing, and Post-Exposure Prophylaxis (PEP).
- **Post-Exposure Prophylaxis (PEP):** PEP for HIV exposure should be initiated as soon as possible, ideally **within 2 hours** of exposure.

- **Bloodborne Transmission Risks:** Risk of HIV transmission after a needle stick is **< 0.3%** (very low). HCV risk is ~1.8%. HBV risk is 6–30% in unvaccinated individuals.
- **Hepatitis B Vaccination & Testing:**
- 3-dose vaccine series. Post-vaccination serologic antibody testing (anti-HBs titer) should be performed **1–2 months after the final injection**.
- **Healthcare Worker Immunizations & Restrictions:**
- **Recommended Vaccines:** Hepatitis B, Influenza, MMR, Varicella, Tdap, COVID-19. (Hepatitis A is NOT bloodborne and NOT routinely required).
- **Work Restrictions:** Required for **active conjunctivitis, measles, mumps, rubella, varicella/shingles, active weeping dermatitis**, and **pediculosis (lice)**. General headaches do NOT require work restrictions.
- **Medical Records:** Employee vaccination/exposure records must be kept confidentially for **duration of employment + 30 years**. (General OSHA injury logs are not required for low-hazard OHC settings unless requested).

[!NOTE] **HIGH-YIELD FOR QUESTIONS 8, 13, 14, 26, 28, 30, 31, 32, 35, 36, 45, 46, 47, 48, 49, 50, 51 & CASE STUDY B:** * After a sharps injury, immediately **wash the wound with soap and water**. * **PEP for HIV** should be initiated **within 2 hours** of exposure. * **Hepatitis B antibody testing** should occur **1–2 months after the 3rd dose**. * Work restrictions are required for **conjunctivitis** and **pediculosis**, but NOT for a **headache**. * Employee exposure records must be maintained for **employment length + 30 years**. * Offices should maintain written **Standard Operating Procedures (SOPs)** for infection control.

7. Question-to-Concept Map (Questions 1 to 51)

Question	Topic Evaluated	Key Fact to Recall
Q1	Regulatory Agencies	OSHA is the federal regulatory body enforcing safety standards.
Q2	Advisory Recommendations	The CDC provides evidence-based recommendations and guidelines.
Q3	Non-Bloodborne Hepatitis	Hepatitis A is transmitted via fecal-oral route (NOT bloodborne).
Q4	Critical Spaulding Item	Dental hygiene cures penetrate tissue and are critical items.
Q5	Semicritical Spaulding Item	Mouth mirrors touch mucous membranes and are semicritical.
Q6	Noncritical Spaulding Item	Blood pressure cuffs touch intact skin and are noncritical.
Q7	Portal of Exit	A cough or sneeze acts as a portal of exit from a source host.

Q8	Work Restrictions	A headache does NOT require work restrictions (conjunctivitis does).
Q9	Engineering Controls	Engineering controls are devices that isolate bloodborne hazards.
Q10	Work Practice Controls	One-handed scoop recapping is a work practice control.
Q11	Active TB Protocol	Patients with active TB must be treated in airborne isolation rooms.
Q12	Standard Precautions	Standard precautions apply to blood, mucosa, and body fluids except sweat .
Q13	Recommended Vaccines	Hepatitis B vaccine is recommended for dental healthcare workers.
Q14	HBV Post-Vaccine Titer	Antibody testing occurs 1–2 months after 3rd Hepatitis B dose .
Q15	Direct Contact Definition	Direct contact involves person-to-person physical contact with blood/fluids.
Q16	Glutaraldehyde Use	Glutaraldehyde is a chemical sterilant, but critical items require heat sterilization.
Q17	DUWL Water Quality	Water used in OHC must contain <= 500 CFU/mL heterotrophic bacteria.
Q18	High-Level Disinfectant	High-level disinfectants kill all microbes including spores (chemical sterilant).
Q19	Intermediate Disinfectant	Intermediate-level disinfectants must carry a tuberculocidal kill claim .
Q20	Spore Exception	Intermediate disinfectants do NOT kill C. difficile spores .
Q21	Clinical Contact Surface	Housekeeping surfaces (floors/walls) are NOT clinical contact surfaces.
Q22	PPE Removal Sequence	Gloves are removed first, followed by eyewear and mask.
Q23	Disinfectant Contact Time	Surface disinfectants must remain wet for manufacturer contact time.

Q24	Sharps Disposal	Sharps containers must be puncture-resistant, leakproof, and labeled.
Q25	Biological Monitoring	Spore testing (biologic monitoring) should be performed weekly .
Q26	Post-Exposure Action	Immediately wash needlestick injury with soap and water.
Q27	Cleaning Method Exception	Hand scrubbing with a short-handled brush is an improper method.
Q28	Post-Exposure PEP Timing	PEP for HIV is best initiated within 2 hours of exposure.
Q29	Hand Hygiene Timing	Wash hands before gloving and immediately after removing gloves.
Q30	HIV Needlestick Risk	The risk of HIV infection after a needle stick is < 0.3% (very low).
Q31	Record Retention	Exposure medical records kept for duration of employment + 30 years .
Q32	OSHA Injury Log	OHC offices are low-hazard and exempt from routine OSHA injury logs.
Q33	Sterilizer Regulation	The FDA regulates safety/effectiveness of sterilizers and ultrasonics.
Q34	Infection Control Goal	Ultimate goal is preventing pathogen transmission in oral healthcare.
Q35	Airborne Diseases	<i>M. tuberculosis</i> and <i>Legionella</i> are transmitted via airborne aerosols.
Q36	Non-Bloodborne Pathogen	Varicella-zoster virus is a respiratory pathogen (not bloodborne).
Q37	Surface Barrier Purpose	Surface barriers prevent contamination of difficult-to-clean surfaces.
Q38	Hand Lotion Components	Use water-based lotions during workday; petroleum lotions degrade latex.
Q39	Employer Responsibility	Employer must pay for post-exposure evaluation and follow-up.

Q40	Carrier Definition	A carrier is recovered/asymptomatic but can transmit pathogens.
Q41	Contaminated Laundry	Handle contaminated laundry with minimal agitation in labeled bags.
Q42	Regulated Medical Waste	Liquid blood, saturated gauze, and sharps are regulated medical waste.
Q43	Instrument Packaging	Packaging maintains sterility after instruments are removed from sterilizer.
Q44	Wicking Phenomenon	Wet instrument packages can draw in microbes via wicking .
Q45	Protocol Standardization	Offices should maintain written Standard Operating Procedures (SOPs) .
Q46	Temporary Staff Protocol	Temporary staff should consult the office SOP manual for protocols.
Q47	Sterilization Monitoring	Physical, chemical, and biological monitors ensure sterilization validity.
Q48	SOP Accessibility	Staff should refer to SOP manual for infection control questions.
Q49	Infection Control Role	Offices should designate an Infection Control Coordinator (ICC) .
Q50	Case A (Legionella)	<i>Legionella</i> transmitted community-to-patient via DUWL aerosols .
Q51	Case B (Needlestick)	Immediate first aid for needlestick is soap and water washing .

8. Clinical Vocabulary (English – Spanish)

Clinical English Term	Spanish Translation	Clinical Definition / Significance
Standard Precautions	Precauciones Estándar	Trata sangre, fluidos corporales, mucosa y piel no intacta como infectados.
Engineering Controls	Controles de Ingeniería	Dispositivos que aíslan/eliminan riesgos (ej. contenedores de punzocortantes).

Work Practice Controls	Controles de Práctica de Trabajo	Comportamientos que reducen riesgos (ej. técnica de encapuchado con una mano).
Critical items	Artículos críticos	Instrumentos que penetran tejido/hueso; requieren esterilización por calor.
Semicritical items	Artículos semicríticos	Instrumentos que tocan mucosas; requieren esterilización o desinfección alta.
Noncritical items	Artículos no críticos	Objetos que tocan piel intacta únicamente (ej. manguito de presión).
Airborne transmission	Transmisión por aire (aerosoles)	Inhalación de microgotas suspendidas en aire (<i>Legionella</i> , <i>M. tuberculosis</i>).
Tuberculocidal claim	Capacidad tuberculocida	Requisito indispensable para desinfectantes de nivel intermedio.
Wicking	Capilaridad (Wicking)	Absorción de microbios a través de empaques de esterilización húmedos.
Legionnaires' disease	Enfermedad del Legionario	Infección pulmonar bacteriana severa por inhalar aerosoles con <i>Legionella</i> .
Post-exposure prophylaxis (PEP)	Profilaxis post-exposición (PPE)	Tratamiento preventivo iniciado preferiblemente dentro de las 2 horas tras exposición.
Anti-HBs titer	Título de anticuerpos anti-HBs	Prueba de laboratorio 1-2 meses post-vacuna de Hepatitis B para verificar inmunidad.
Pediculosis	Pediculosis (piojos)	Infestación cutánea que requiere restricción laboral del profesional dental.
Clinical contact surface	Superficie de contacto clínico	Superficie de alto contacto en área de tratamiento que requiere barrera o desinfección.
Housekeeping surface	Superficie de limpieza general	Pisos y paredes que requieren limpieza básica con agua/detergente.

9. Rapid Recall High-Yield Facts

- **OSHA** → Regulatory body enforcing workplace safety; **CDC** provides advisory recommendations.
- **FDA** → Regulates safety and effectiveness of sterilizers, ultrasonics, and medical devices.

- **Critical Items** → Penetrate soft tissue/bone (**curettes, scalers**) → **Must be heat sterilized**.
- **Semicritical Items** → Touch mucous membranes (**mouth mirrors, X-ray sensors**).
- **Noncritical Items** → Touch intact skin only (**blood pressure cuff**).
- **Standard Precautions** → Apply to blood, **all body fluids except sweat**, mucous membranes, and nonintact skin.
- **Active Tuberculosis** → Patients **MUST** be treated in an **airborne isolation room**.
- **Engineering Controls** → Sharps containers, self-sheathing needles; **Work Practice Controls** → One-handed scoop technique.
- **Intermediate Disinfectants** → Must carry a **tuberculocidal claim**; do NOT kill *C. difficile* spores.
- **DUWL Biofilm (*Legionella*)** → Causes airborne **Legionnaires' disease** via water vapor inhalation.
- **DUWL Quality Standard** → Dental unit water must contain **≤ 500 CFU/mL** heterotrophic bacteria.
- **Sharps Exposure Protocol** → **Wash immediately with soap and water**; do NOT scrub or squeeze.
- **Post-Exposure PEP Timing** → HIV PEP should be initiated **within 2 hours** of occupational exposure.
- **Employer Exposure Obligation** → Must provide confidential medical evaluation and PEP at **zero cost to employee**.
- **HBV Post-Vaccine Titer** → Test anti-HBs antibody levels **1–2 months after 3rd dose**.
- **Record Retention** → Employee exposure records kept for **employment duration + 30 years**.
- **Wicking** → Environmental contamination drawn into wet sterilizer packages.
- **Work Restrictions** → Required for **conjunctivitis** and **pediculosis**; NOT required for headaches.